



# The design, analysis, and cost estimation of a generic adder and subtractor using the layered T (LT) logic reduction methodology with a quantum-dot cellular-automata-based approach

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## Abstract

The quantum-dot cellular automata (QCA) is considered to be one of the ground-breaking nanotechnologies developed over the last two decades. A layered T (LT) logic cell library is constructed herein, and the methodology is extended to generic adder and subtractor module designs. The two proposed algorithms lead to more efficient QCA layout designs for an  $n$ -bit ripple carry adder (RCA) and subtractor based on an effective clock zone assignment approach. The suggested one-, four-, and eight-bit RCAs and subtractors surpass most of their existing counterparts by offering lower effective area and cell complexity. A comparative analysis is presented regarding the complexity, irreversible power dissipation, and  $\text{Cost}_\alpha$  of the proposed  $n$ -bit layouts from a cost estimation purview.

**Keywords** Quantum-dot cellular automata (QCA) · Layered T gate · Adder and subtractor · Complexity · Irreversible power dissipation (IPD) ·  $\text{Cost}_\alpha$

## 1 Introduction

The continuous downscaling of complementary metal–oxide–semiconductor (CMOS) dimensions results in severe performance issues such as quantum effects, charge quantization, leakage current, increased device fabrication costs, and many more [1]. It thus become imperative for researchers to develop efficient alternative technologies, based for example on carbon nanotubes [2], spintronics [3], tunneling phase logic [4], molecular electronics [5], and quantum-dot cellular automata (QCA) [6] to mitigate such limitations. The QCA, which remains in focus due to its ultrasmall feature sizes and extremely high operating speeds, has become a promising approach to link cellular

automata and quantum systems for the realization of computational blocks for nanoprocessors. Most importantly, the QCA has almost zero power dissipation during operation [7]. This emerging nanotechnology, which relies on quantum-mechanical tunneling phenomena, can help to realize combinational and sequential logic devices [8].

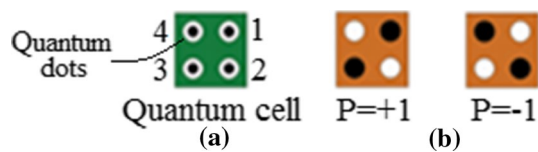
QCA architectures are carefully built by placing QCA cells and allocating a specific clock cycle. A QCA cell consists of four quantum dots located at the four corners of the top surface of a molecular beam epitaxy (MBE)-grown gallium arsenide/aluminum gallium arsenide (GaAs/AlGaAs) heterostructure (Fig. 1a). When two electrons are added to such a cell, they settle at any two diagonal quantum dots due to the columbic (repulsive) force between them. These pairs of diagonal quantum dots form two antipodal states, either diagonally right or diagonally left (Fig. 1b). If the diagonal right antipodal state is occupied, then the QCA cell exhibits polarization  $P = +1$ . Alternatively, the cell exhibits polarization  $P = -1$  when the diagonal left antipodal state is adopted. The polarization values  $P = +1$  and  $-1$  specify logic states 1 and 0, respectively. A group of coherent QCA cells can interact and transmit information from one location to another. In this regard, the primary logic functions that can be instantiated using QCA are the majority voter, binary wire, and inverter. A majority voter can function as

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**Fig. 1** **a** A quantum cell with polarization. **b** The states corresponding to  $P = +1$  and  $P = -1$

either an AND or an OR gate, and a designer can construct any digital logic circuits by combining such gates, wires, and inverters. The clock cycles drive the logic flow inside a QCA layout [9].

Although such QCAs have been characterized and tested, and their behavior synthesized, several areas of QCA research remain unexplored. Indeed, researchers continue to seek a better understanding of the thermal behavior and fabrication difficulties of QCA devices [10], and the possibility of implementing QCA digital circuits. Existing logic reduction methodologies cannot efficiently use QCA design rules to generate the layouts of combinational circuits using QCA [11]. Moreover, current methods also cannot provide specific guidelines to explore the design of generic modules.

In this work, the layered T logic reduction technique [12] is employed to facilitate the following:

1. The creation of an LT logic library
2. The design and implementation of lower-order combinational modules for an adder and subtractor, using the cells of the proposed LT library
3. A proposal for algorithms to extend and reuse these lower-order modules in a corresponding higher-order adder and subtractor via efficient use of the QCA design rules [11]
4. A cost analysis [13] and performance evaluation of the proposed designs in the presence of a generic input

Four types of implementations are available to achieve QCAs: semiconductor [14], metal dot [15], molecule [16], and magnetic [17]. Both semiconductor and metal-dot QCAs operate at cryogenic temperatures. However, molecular and magnetic QCA can function at room temperature. Here, a semiconductor QCA-based architecture using gallium

arsenide (GaAs) is proposed to simulate the layouts in the computer-aided design tool QCADesigner 2.0.3 [18]. Moreover, this approach could also be tested and verified using other types of QCA.

The remainder of this manuscript is organized as follows: Sect. 2 presents the background of QCA, where the primary QCA logic gates are described. The logic reduction methodologies associated with these logic gates are also discussed in this section. Moreover, this section reviews the relevant works on QCA. Section 3 describes the LT logic reduction technique. In this section, an LT gate library is constructed using the appropriate characteristic equations and QCA layouts. Section 4 introduces the algorithms that are used to generate the generic adders and subtractors. Additionally, the cost function is described, accompanied by a comparative analysis of the proposed QCA layouts versus existing counterparts. Finally, Sect. 5 concludes this work.

## 2 QCA background and related works

### 2.1 QCA background

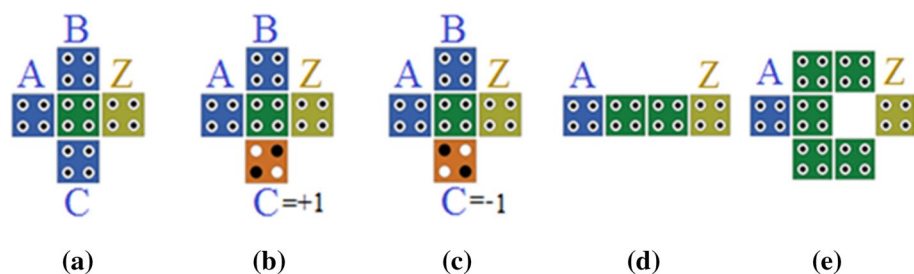
The commonly available logic reduction methods in QCA are the majority voter (MV), AND–OR–INVERT (AOI), NAND–NOR–INVERT (NNI), universal QCA logic gate (UQCALG), FNZ, and layered T (LT) gate.

MV logic, which is the elementary gate in QCA, is constructed from five QCA cells as shown in Fig. 2a. The MV has three inputs (A, B, and C) and computes one output (Z) as shown in Eq. (1):

$$Z = AB + BC + CA \quad (1)$$

A two-input AND or OR gate can be built by permanently polarizing input cell C to  $+1$  or  $-1$ , as specified in Fig. 2b and c, respectively. Figure 2d presents an array of QCA cells that carry information from the input to output cells. Such an arrangement is termed a binary wire in QCA. The conventional inverter is formed by adequately positioning seven QCA cells as shown in Fig. 2e [19].

**Fig. 2** **a** A majority voter, with polarization **b**  $P = +1$  and **c**  $P = -1$ , **d** a binary wire, and **e** an inverter



The AOI gate was introduced [20] to map logic synthesis tools more efficiently than when using the MV. It has five inputs (A, B, C, D, and E) and computes one output (Z) as shown in Eq. (2):

$$Z = DE + (D + E)(\overline{AC} + \overline{AB} + \overline{BC}) \quad (2)$$

Permanently polarizing any two of these five inputs forces the AOI gate to operate explicitly as a one-level or two-level gate. The NNI gate [21], which is implemented as an inverted AOI gate, can optimize the effective area and improve the delay of QCA layouts. The NNI gate receives five inputs (A, B, C, D, and E) and produces one output (Z) as shown in Eq. (3):

$$Z = \overline{DE} + (\overline{D} + \overline{E})(AB + BC + CA) \quad (3)$$

The three-input four-output UQCALG employs coupled majority voter minority logic (CMVMIN) to establish an output as shown in Eq. (4) [22]:

$$\begin{cases} Z1 = (\overline{A} + \overline{B}) \\ Z2 = \overline{AB} + \overline{AC} \\ Z3 = \overline{AB} + \overline{AC} \\ Z4 = A + B \end{cases} \quad (4)$$

The UQCALG represents a novel paradigm in QCA logic design, but the optimization of such layouts remains challenging in terms of effective area and delay due to its high area utilization factor [23]. The FNZ gate is instantiated using eight QCA cells. It takes three inputs (A, B, and C) and yield one output (Z) as shown in Eq. (5) [24]:

$$Z = \overline{AB} + (A \oplus B) \quad (5)$$

Figure 3a, b, c, and d show the AOI, NNI, UQCALG, and FNZ gate, respectively.

We proposed universal NAND, NOR, and exclusive OR (EX-OR) functions using layered T (LT) logic in Refs. [12, 25, 26]. Here, we build an LT logic library in that includes

all the elementary logic gates. The logical synthesis of the LT library using the QCADesigner tool reveals that the LT designs are more suitable and flexible than their MV counterparts. The LT logic library is further used to construct lower- and higher-order adder and subtractor modules in subsequent sections.

## 2.2 Related works

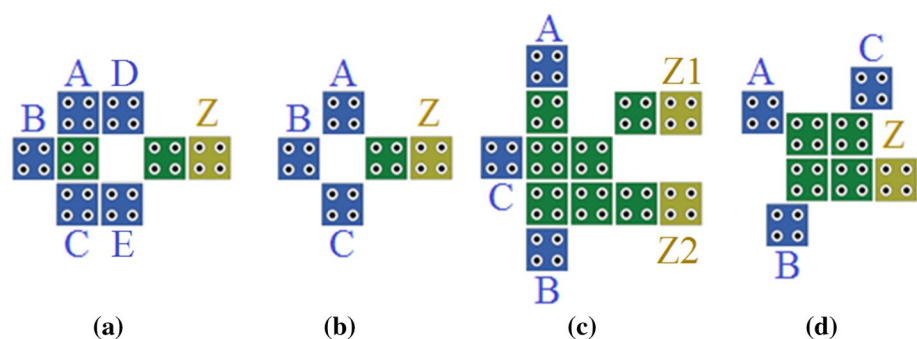
Researchers are working on various problems and solutions associated with the design of adders and subtractors. Although the drawbacks and loopholes of existing designs still require further exploration, no such complete and systematic studies that address the issues related to the above-mentioned combinational logic modules exist. A few attempts are available, reporting systematic literature studies of adders, multiplexers, and memory designs [27–29]. However, these studies are not adequate and compact. This section highlights the works related to adders and subtractors and explores the key areas requiring further work. Table 1 summarizes the existing adder and subtractor designs [30–53] on the basis of (i) the type of crossover used, (ii) the type of logic reduction applied, (iii) their designability and extendability, and (iv) any issues related to the design rules. Table 1 presents a review of the design techniques used for existing adders and subtractors and comments on the issues with the mentioned approaches.

According to Table 1, 91.6% of such work has focused on the adder module, while the others have considered subtractor designs. Significantly, 50% of the adder and 100% of the subtractor QCA layouts suffer from issues related to the design rules. Although logical and multilayer crossovers are preferred, only 8%, 33%, and 25% of the layouts use logical, multilayer, and coplanar crossovers, respectively. Moreover, 83.3%, 41.6%, and 12.5% of the layouts are redesignable and extendable to 1-bit, 4-bit, and 8-bit counterpart designs.

One can thus highlight the following points regarding existing adder and subtractor designs:

1. Most of the existing modules are not extendable to higher-order counterparts.

**Fig. 3** The QCA layouts of the **a** AOI, **b** NNI, **c** CMVMIN, and **d** FNZ gate



**Table 1** A summary of existing adders and subtractors

| No | Adder and subtractor design       |                                                                                                                                       |
|----|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Type of module                    | Adder [30–53]<br>Subtractor [37, 47]                                                                                                  |
| 2  | Type of crossover used            | Coplanar [30, 32, 33, 38, 51, 52]<br>Multilayer [31, 35, 39, 43, 47–50]<br>Logical [40, 52]<br>Not used [34, 36, 41, 42, 44–46, 53]   |
| 3  | Type of logic reduction used      | Three-input MV [30–53]<br>Five-input MV [39, 43, 48]<br>Coupled three-input and five-input MVs [42, 44, 45, 50]                       |
| 4  | Redesignability and extendability | 1 bit [30–53]<br>4 bit [30, 31, 35, 36, 40, 42, 43, 48–50]<br>8 bit [37, 49, 52]<br>16 bit [31, 49]<br>32 bit [40]<br>64 bit [31, 40] |
| 5  | Issues related to design rules    | Present [34, 35, 37, 40–43, 47–52]<br>Not present [30–33, 36, 38–40, 44–46]                                                           |

- Although logical and multilayer crossovers are efficient [54], most of the modules do not use them.
- Design rules are not adequately followed in QCA layouts.
- Detailed cost estimation is not carried out in existing works.

We present herein the LT logic reduction technique and then instantiate an adder and subtractor by using LT logic to mitigate the above-mentioned drawbacks.

### 3 The layered T (LT) logic reduction technique

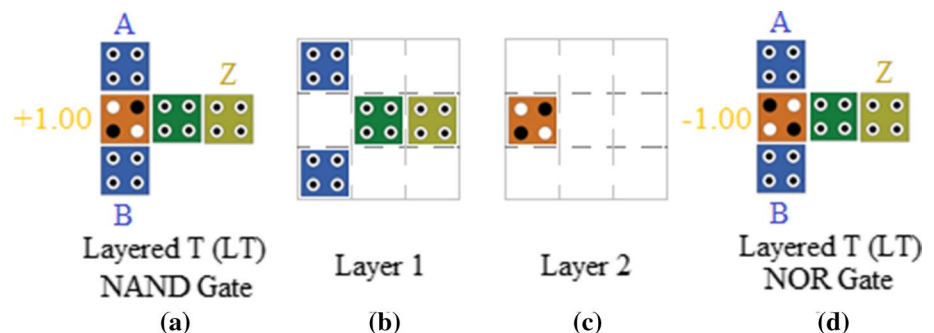
The layered T (LT) logic reduction methodology involves the creation of LT cell libraries, as described below:

#### 3.1 LT NAND and NOR gates

An LT NAND gate has two inputs (A and B) and one output (Z) as shown in Fig. 4a. Four QCA cells are arranged on layer 1, and one QCA cell with  $P = +1$  is placed at layer 2, as shown in Fig. 4b and c, respectively. Layer 2 is situated vertically above layer 1. The output P exhibits stable energy levels when the cell dimension and dot diameter vary from 14 to 22 nm. It is 4 nm to 6 nm, respectively, with a minimum layer separation of 42 nm. However, we use the specific cell parameters presented in Table 2 to test the LT gates. The QCA layout of the LT NOR gate is shown in Fig. 4d. If the polarization of the cell in layer 2 is fixed at  $-1$ , then the layout operates as a NOR. Equation 6 represents the mathematical expression for the LT NAND and NOR gates.

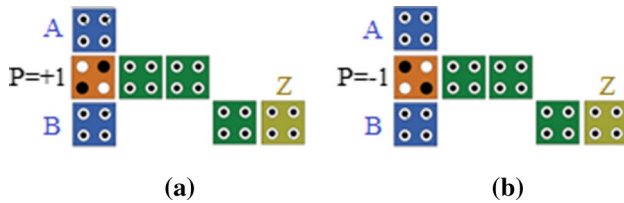
$$\begin{cases} Z = L_T^+(A, B) = \overline{AB}, & \text{if layer 2 cell logic is 1} \\ Z = L_T^-(A, B) = \overline{A + B}, & \text{if layer 2 cell logic is 0} \end{cases} \quad (6)$$

**Fig. 4** **a** The QCA layout of the LT NAND gate, showing **b** layer 1 and **c** layer 2. **d** The QCA layout of the LT NOR gate



**Table 2** The parameter specifications used in QCADesigner

| No. | Parameter        | Value or range (in nm) |
|-----|------------------|------------------------|
| 1   | Cell dimension   | 18 × 18                |
| 2   | Layer separation | 42                     |
| 3   | RoE              | 65                     |
| 4   | Cell spacing     | 2                      |
| 5   | Dot diameter     | 4                      |

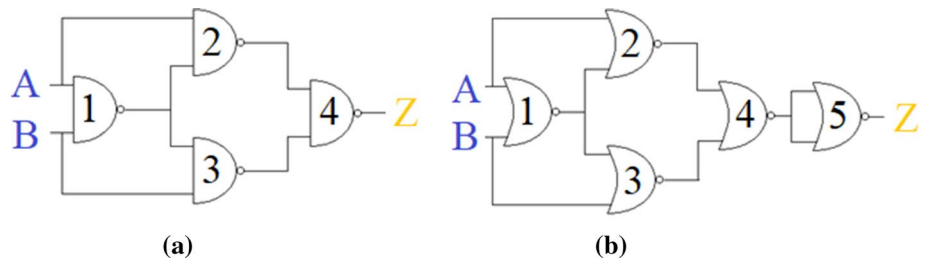
**Fig. 5** The QCA layouts of the LT **a** AND and **b** OR gates

### 3.2 The LT AND and OR gates

The QCA layouts of the LT AND and OR gates are shown in Fig. 5a and b, respectively. Six QCA cells are placed in layer 1 of the layout, while one fixed-polarization cell is placed in layer 2. If we suppose that the polarization of the cell in layer 2 becomes +1, then this configuration functions as an AND gate. Similarly, if it is fixed to −1, then the block operates as an OR gate.

Equation (7) presents the mathematical expressions for the LT AND and OR gates with the respective functions of  $L_{TA}^+$  and  $L_{TO}^-$ .

$$\begin{cases} Z = L_{TA}^+(A, B) = AB, & \text{if layer 2 cell logic is 1} \\ Z = L_{TO}^-(A, B) = A + B, & \text{if layer 2 cell logic is 0} \end{cases} \quad (7)$$

**Fig. 6** The construction of an Ex-OR gate by using **a** NAND or **b** NOR gates

### 3.3 The LT Ex-OR and Ex-NOR gates

Ex-OR is a two-literal operation, having significant applications in digital arithmetic circuits. It generates a logic 1 for dissimilar logic at the inputs; otherwise, it produces a logic 0. The logical expression for the Ex-OR gate is given by Eq. (8).

$$Z = A\bar{B} + \bar{A}B \quad (8)$$

An Ex-OR gate can be realized by using either four NAND or five NOR gates, as shown in Fig. 6a and b, respectively.

We use  $L_{TEx}()$  function to denote the LT EX-OR logically. The NAND realization is chosen to minimize the gate count, optimize the effective area of the QCA layout, and reduce the O-Cost. If four LT NAND gates are placed at positions 1, 2, 3, and 4 of Fig. 6a, then the  $L_{TEx}$  gate can be instantiated. The LT NAND is placed at position 1, generating  $L_T^+(A, B) = \bar{A}\bar{B}$  at its output, which further feeds the LT counterparts placed at positions 2 and 3.

Thus, LT NAND 2 and NAND 3 receive two input pairs of  $A, L_T^+(A, B)$  and  $B, L_T^+(A, B)$ , respectively. In the final stage, LT NAND 4 evaluates the intermediary output, temp, to generate the outcome,  $Z$ , as shown in Eqs. 9 and 10 respectively.

$$\text{temp} = (L_T^+(L_T^+(A, L_T^+(A, B)), L_T^+(B, L_T^+(A, B))) \quad (9)$$

$$Z = \overline{(A, \bar{A}\bar{B})(B, \bar{A}\bar{B})} = \bar{A}\bar{B} + \bar{A}B \quad (10)$$

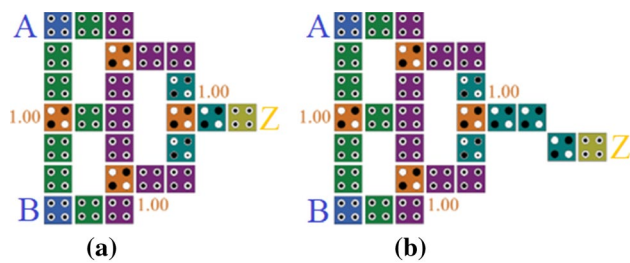
The equation for the  $L_{TEx}$  gate is given in Eq. (11).

$$Z = A \oplus B = L_{TEx}(A, B) = L_T^+(L_T^+(A, L_T^+(A, B)), L_T^+(B, L_T^+(A, B))) \quad (11)$$

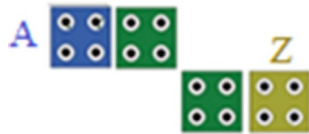
An  $L_{TExn}$  gate can be constructed similarly, as discussed in the description of the  $L_{TEx}$  gate. An  $L_{TEx}$  gate followed by an additional inverter generates the  $L_{TExn}$  gate. The  $L_{TExn}()$  is used to create  $AB + \bar{A}\bar{B}$ , as follows in Eq. (12):

$$Z = A \odot B = L_{TExn}(A, B) = \overline{L_T^+(L_T^+(A, L_T^+(A, B)), L_T^+(B, L_T^+(A, B)))} \quad (12)$$





**Fig. 7** The QCA layouts of the **a** LTeX and **b** LTExn gates



**Fig. 8** The QCA layout of the LT inverter gate

Figures 7a and b show the layouts of the LTeX and LTExn gates.

When using the LT logic reduction technique, inversion is implemented by using the four-cell arrangement shown in Fig. 8. The equation for LT inversion is represented mathematically by Eq. (13).

$$Z = L'_T(A) = \bar{A} \quad (13)$$

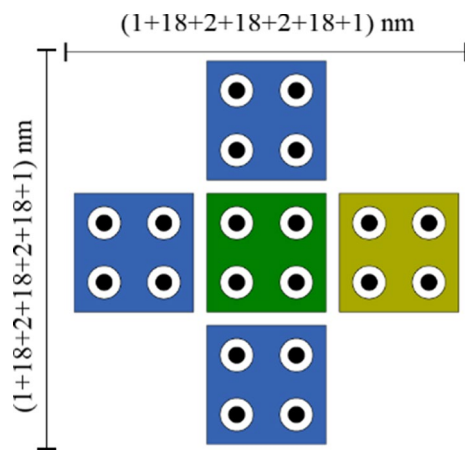
We summarize the logical symbols, collate the LT equations, and report the O-Cost, effective area, and delay requirements to implement these LT gates in Table 3.

We denote the effective area of the MV by  $A_{MV}$ , in terms of which the effective area of each LT gate can be expressed. This diagnostic comparison can help nanoelectronics researchers to choose a suitable logic alternative for nano central processing unit (nano CPU) design [55]. If the cell spacing is 2 nm, and the length and breadth of a QCA cell are fixed respectively at 18 nm, then the active area of a three-input MV becomes  $(60 \times 60) \text{ nm}^2 = 3600 \text{ nm}^2$ . The details of the length and breadth of the MV area is shown in Fig. 9. Therefore, the value of  $A_{MV}$  becomes  $3600 \text{ nm}^2$ . If the area of the LT NOT gate is  $2964 \text{ nm}^2$ , then it becomes  $0.82A_{MV}$ . Similarly, if the LT NAND consumes an effective area of  $3364 \text{ nm}^2$ , then the calculation shows that it is merely  $0.93A_{MV}$ .

The area requirements for the remaining LT counterparts in terms of  $A_{MV}$  are summarized in Table 4. The average area requirements of the logic cells using the LT methodology consume  $2.65A_{MV}$ , which is 29.52% lower than for the MV counterparts, as summarized in Table 4.

**Table 3** A design summary of the one- and two-input LT gate library

| No | Gate   | Symbol | LT equation     | O-Cost # | Effective area (nm <sup>2</sup> ) | QCA clock cycle |
|----|--------|--------|-----------------|----------|-----------------------------------|-----------------|
| 1  | NOT    |        | $L'_T(A)$       | 4        | 2964                              | 0.25            |
| 2  | NAND   |        | $L_T^+(A,B)$    | 5        | 3364                              | 0.25            |
| 3  | AND    |        | $L_{TA}^+(A,B)$ | 7        | 5684                              | 0.25            |
| 4  | NOR    |        | $L_T^-(A,B)$    | 5        | 3364                              | 0.25            |
| 5  | OR     |        | $L_{TO}^-(A,B)$ | 7        | 5684                              | 0.25            |
| 6  | Ex-OR  |        | $L_{TX}(A,B)$   | 26       | 19,044                            | 0.75            |
| 7  | Ex-NOR |        | $L_{TXn}(A,B)$  | 28       | 24,564                            | 0.75            |



**Fig. 9** The detailed dimensions of a three-input MV

#### 4 Combinational logic module designs using the LT logic reduction technique

This section introduces a way to generate QCA layouts by using the LT cell library with efficient clocking zone assignments. The difficulties regarding QCA designs related to

the adder and subtractor are recognized. A comprehensive solution with parameter estimation is provided to achieve optimal QCA design metrics.

##### 4.1 Adder design

Different architectures are possible to implement adders, including the ripple carry adder (RCA), the carry look-ahead adder (CLA), the carry-skip adder (CSkipA), the carry-propagation adder (CPA), the carry-select adder (CSeIA), and the parallel-prefix adder (PPA) [13]. The CLA can optimize the propagation delay by examining all the available input bits and producing carry bits for all the stages simultaneously. This avoids the ripple carry delay and speeds up the addition process [56]. Our main aim here is thus to define how to generate a CLA. Algorithm 1 is proposed to generate higher-order LT RCA with inputs  $A = (A[0]A[1] \dots A[n-1])$ ,  $B = (B[0]B[1] \dots B[n-1])$  and  $C = (C[0]C[1] \dots C[n-1])$ , and the outputs, sum and carry, as  $S = (S[0]S[1] \dots S[n-1])$  and  $CO = (CO[0]CO[1] \dots CO[n-1])$ , respectively. We assume the initial carry input as  $Cin$ .

**Algorithm 4.1: The n-bit LT RCA generation (where  $n=2^m$ ,  $m=0, 1, 2, \dots$ )**

INPUT:  $A=(A[0]A[1] \dots A[n-1])$ ,  $B=(B[0]B[1] \dots B[n-1])$  and  $C=(C[0]C[1] \dots C[n-1])$  in binary and  
OUTPUT:  $Sum=(S[0]S[1] \dots S[n-1])$  and  $CO=(CO[0]CO[1] \dots CO[n-1])$  assuming initial one-bit carry input as  $Cin$ .

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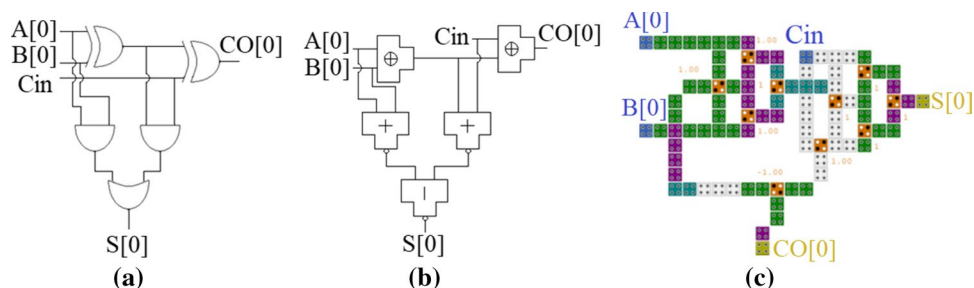
1.  $i=0$  //Initialize loop index
2.  $S[0]=LTeX(A[0], LTeX(B[0], Cin));$  //Computation of initial values
    $CO[0]=L\tau_o(L_{TA}^+(A[0], B[0]), L_{TA}^+(Cin, LTeX(A[0], B[0])));$ 
3. for  $i=1$  to  $n-1$  //loop
   do
      $S[i]=LTeX(A[i], LTeX(B[i], CO[i-1]));$ 
      $CO[i]=L\tau_o(L_{TA}^+(A[i], B[i]), L_{TA}^+(CO[i-1], LTeX(A[i], B[i])));$ 
4. end //end of the loop

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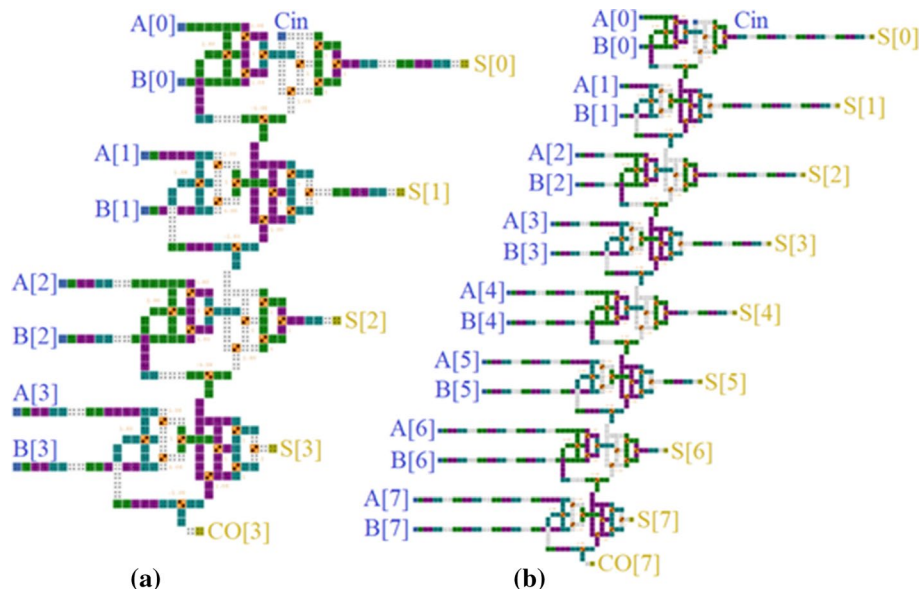
**Table 4** The effective area requirements of the LT cells relative to the  $A_{MV}$  values

| No      | Logic gate | Effective area in $A_{MV}$ ( $nm^2$ ) when implemented using LT | Effective area in $A_{MV}$ ( $nm^2$ ) when implemented using MV |
|---------|------------|-----------------------------------------------------------------|-----------------------------------------------------------------|
| 1       | NOT        | $0.88A_{MV}$ (2964)                                             | $1.25A_{MV}$ (4524)                                             |
| 2       | NAND       | $0.93A_{MV}$ (3364)                                             | $2.22A_{MV}$ (8004)                                             |
| 3       | AND        | $1.67A_{MV}$ (5684)                                             | $A_{MV}$ (3600)                                                 |
| 4       | NOR        | $1A_{MV}$ (3364)                                                | $2.22A_{MV}$ (8004)                                             |
| 5       | OR         | $1.67A_{MV}$ (5684)                                             | $A_{MV}$ (3600)                                                 |
| 6       | Ex-OR      | $5.44A_{MV}$ (19,044)                                           | $8.69A_{MV}$ (31,284)                                           |
| 7       | Ex-NOR     | $7A_{MV}$ (24,564)                                              | $9.94A_{MV}$ (35,808)                                           |
| Average |            | $2.65A_{MV}$                                                    |                                                                 |

**Fig. 10** The **a** logic diagram, **b** LT schematic, and **c** QCA layout of the one-bit LT RCA



**Fig. 11** The **a** four- and **b** eight-bit LT RCA layouts



A one-bit LT RCA block is implemented by utilizing algorithm 1, then we use an algorithm to generate four- and eight-bit LT RCA counterparts. A one-bit RCA takes three inputs,  $A[0]$ ,  $B[0]$ , and  $Cin$ , and processes sum and carries as shown in Eq. (14).

$$\begin{cases} S[0] = A[0] \oplus B[0] \oplus Cin \\ CO[0] = A[0]B[0] + B[0]Cin + CinA[0] \end{cases} \quad (14)$$

The one-bit RCA requires two Ex-OR gates, two AND gates, and one OR gate, as shown in Fig. 10a. The LT equation for a one-bit LT RCA is given in Eq. (15).

$$\begin{cases} S[0] = L_{TO}(L_{TA}^+(A[0], L_{TE}(B[0], Cin)) \\ CO[0] = L_{TO}^-(L_{TA}^+(A[0], B[0]), L_{TA}^+(Cin, L_{TE}(A[0], B[0]))) \end{cases} \quad (15)$$

The LT schematic and its QCA layout are shown in Fig. 10b and c, respectively.

Algorithm 1 is used to create higher-order RCA instantiations of four- and eight-bit LT RCA counterparts, as shown in Fig. 11a and b, respectively. The one-, four-, and eight-bit LT RCA counterparts consume 0.119, 1.15, and 4.36  $\mu m^2$

at an O-Cost of 94, 434, and 1058 outputs after 1.5, 3, and 5 clock cycles, respectively. Table 5 presents a comparative analysis of the LT RCA with existing counterparts in terms of the effective area in  $\mu m^2$ , O-Cost, and delay.

## 4.2 Subtractor design

Subtraction is another computational operation that finds applications when calculating the difference between any two numbers. A one-bit full subtractor (FS) subtracts a one-bit subtrahend  $B$  from a one-bit minuend  $A$ , considering the borrow  $BI$  from that specific stage used for the subtraction process in the preceding step. This yields results known as difference and borrow, as given in Eq. (16).

$$\begin{cases} D = A \oplus B \oplus Bin \\ BO = \overline{A}B + \overline{A}Bin + BBin \end{cases} \quad (16)$$

Algorithm 2 explains the generation of the higher-order LT FS with inputs  $A = (A[0]A[1] \dots A[n-1])$  and  $B = (B[0]B[1] \dots B[n-1])$  and outputs  $D = (D[0]D[1] \dots D[n-1])$  and  $BO = (BO[0]BO[1] \dots BO[n-1])$ . We assume an initial borrow input as  $Bin$ . A one-bit LT FS has three inputs  $A[0]$ ,



**Table 5** A comparative analysis of the one-, four-, and eight- LT RCAs with existing designs

| No | Adder design     |                      | Effective area (in nm <sup>2</sup> ) | O-Cost # | QCA clock cycle |
|----|------------------|----------------------|--------------------------------------|----------|-----------------|
| 1  | One-bit LT RCA   | [30] <sup>*</sup>    | 0.16004                              | 145      | 1.25            |
| 2  |                  | [31]                 | 0.177684                             | 137      | 1.25            |
| 3  |                  | [32] <sup>*</sup>    | 0.405764                             | 250      | 2.75            |
| 4  |                  | [33]                 | 0.761600                             | 171      | 1               |
| 5  |                  | [34] <sup>@, *</sup> | 0.548800                             | 131      | 2               |
| 6  |                  | [35] <sup>@</sup>    | 0.321408                             | 74       | 1               |
| 7  |                  | [37] <sup>@, *</sup> | 0.07                                 | 69       | 1               |
| 8  |                  | [38]                 | 0.761600                             | 191      | 1.25            |
| 9  |                  | [39]                 | 0.321408                             | 147      | 0.75            |
| 10 |                  | [40] <sup>@</sup>    | 0.043                                | 59       | 1               |
| 11 |                  | [41] <sup>@</sup>    | 0.06                                 | 80       | 0.75            |
| 12 |                  | [42] <sup>@</sup>    | 0.06                                 | 71       | 1.25            |
| 13 |                  | [43] <sup>@</sup>    | 0.01                                 | 23       | 0.75            |
| 14 |                  | [44] <sup>@</sup>    | 0.04                                 | 41       | 0.5             |
| 15 |                  | [46] <sup>@</sup>    | 0.02                                 | 29       | 0.5             |
| 16 |                  | [47] <sup>@</sup>    | 0.6                                  | 90       | 3               |
| 17 |                  | [48] <sup>@</sup>    | 0.02                                 | 29       | 0.5             |
| 18 |                  | [49] <sup>@</sup>    | 0.01                                 | 31       | 0.75            |
| 19 |                  | [50] <sup>@</sup>    | 0.01                                 | 28       | 0.5             |
| 20 |                  | [51] <sup>@</sup>    | 0.06                                 | 44       | 1.25            |
| 21 |                  | [52] <sup>@</sup>    | 0.027                                | 39       | 1               |
| 22 |                  | [53] <sup>@</sup>    | 0.0924                               | 114      | 1               |
| 23 |                  | This work            | 0.119                                | 94       | 1.5             |
| 24 | Four-bit LT RCA  | [30] <sup>*</sup>    | 0.2109                               | 580      | 1.25            |
| 25 |                  | [31]                 | 3.136                                | 1758     | 3.75            |
| 26 |                  | [35] <sup>@</sup>    | 1.86                                 | 1838     | 2.5             |
| 27 |                  | [36]                 | 1.202                                | 651      | 4.25            |
| 28 |                  | [40] <sup>@</sup>    | 0.208                                | 262      | 1.75            |
| 29 |                  | [42] <sup>@</sup>    | 1                                    | 442      | 2               |
| 30 |                  | [43] <sup>@</sup>    | 0.20                                 | 102      | 1               |
| 31 |                  | [48] <sup>@</sup>    | 0.37                                 | 269      | 3.5             |
| 32 |                  | [49] <sup>@</sup>    | 0.12                                 | 166      | 1.5             |
| 33 |                  | [50] <sup>@</sup>    | 0.06                                 | 135      | 1.25            |
| 34 | Eight-bit LT RCA | This work            | 1.15                                 | 434      | 3               |
| 35 |                  | [36]                 | 3.56                                 | 1499     | 8.25            |
| 36 |                  | [40] <sup>@</sup>    | 0.492                                | 572      | 2.75            |
|    |                  | [42] <sup>@</sup>    | 5                                    | 1254     | 4               |
| 37 |                  | [49] <sup>@</sup>    | 0.40                                 | 430      | 2.5             |
| 38 |                  | This work            | 4.36                                 | 1058     | 5               |

NA, not applicable

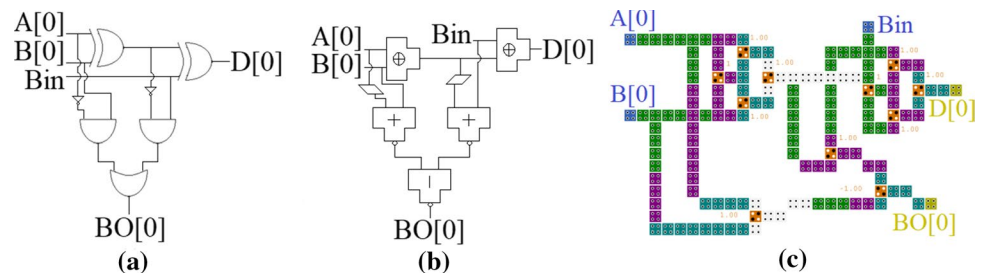
<sup>@</sup> Severe issues with design rules<sup>\*</sup> High-order adder designs are not available

B[0], and Bin and generates the difference and borrow, as provided in Eq. (17), consecutively.

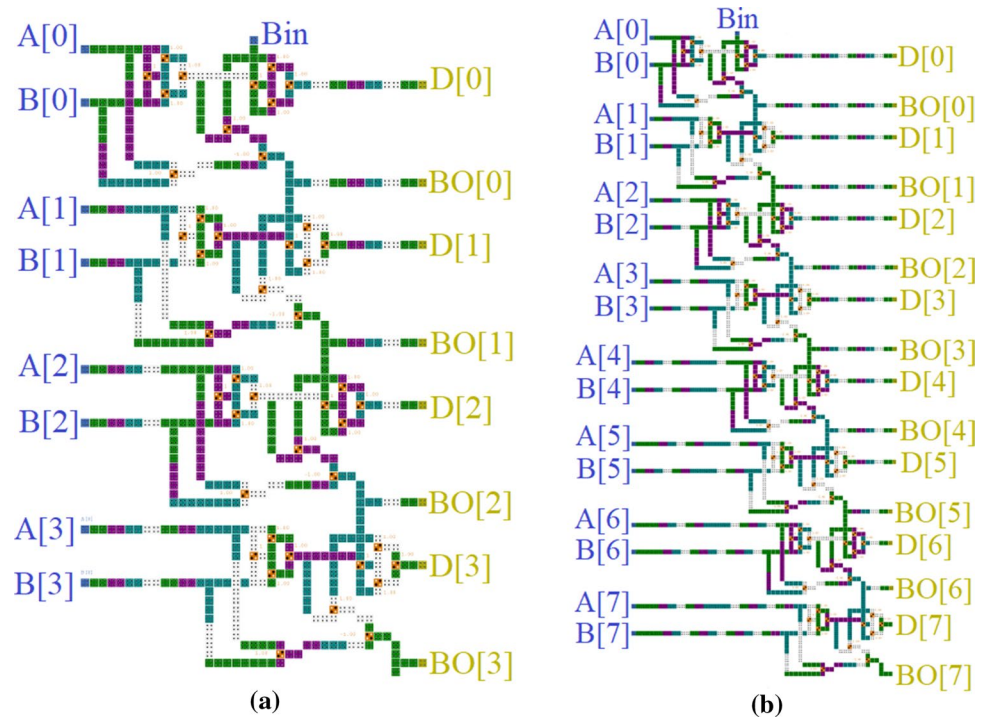
$$\begin{cases} D[0] = A[0] \oplus B[0] \oplus \text{Bin} \\ BO[0] = \overline{A[0]}B[0] + \overline{A[0]}B[0] + B[0]\text{Bin} \end{cases} \quad (17)$$

The LT equation of the one-bit LT FS is generated by utilizing Algorithm 2 as follows in Eq. (18). The LT schematic

**Fig. 12** The **a** block diagram, **b** LT schematic, and **c** QCA layout of the one-bit LT FS



**Fig. 13** The **a** four- and **b** eight-bit LT FS QCA layouts



**Table 6** A comparative analysis of the eight-bit LT RCA with existing designs

| No | LT FS design | Effective area<br>(in nm <sup>2</sup> ) | O-Cost <sup>#</sup> | QCA clock cycle |
|----|--------------|-----------------------------------------|---------------------|-----------------|
| 1  | [37]         | 10.47                                   | 5786                | 27.25           |
| 2  | [52]         | 2.83                                    | 1884                | 28              |
| 3  | This work    | 3.455487                                | 1661                | 5.25            |

**Table 7** The estimated parameters of the proposed adder and subtractor layouts in terms of generic or  $n$ -bit input

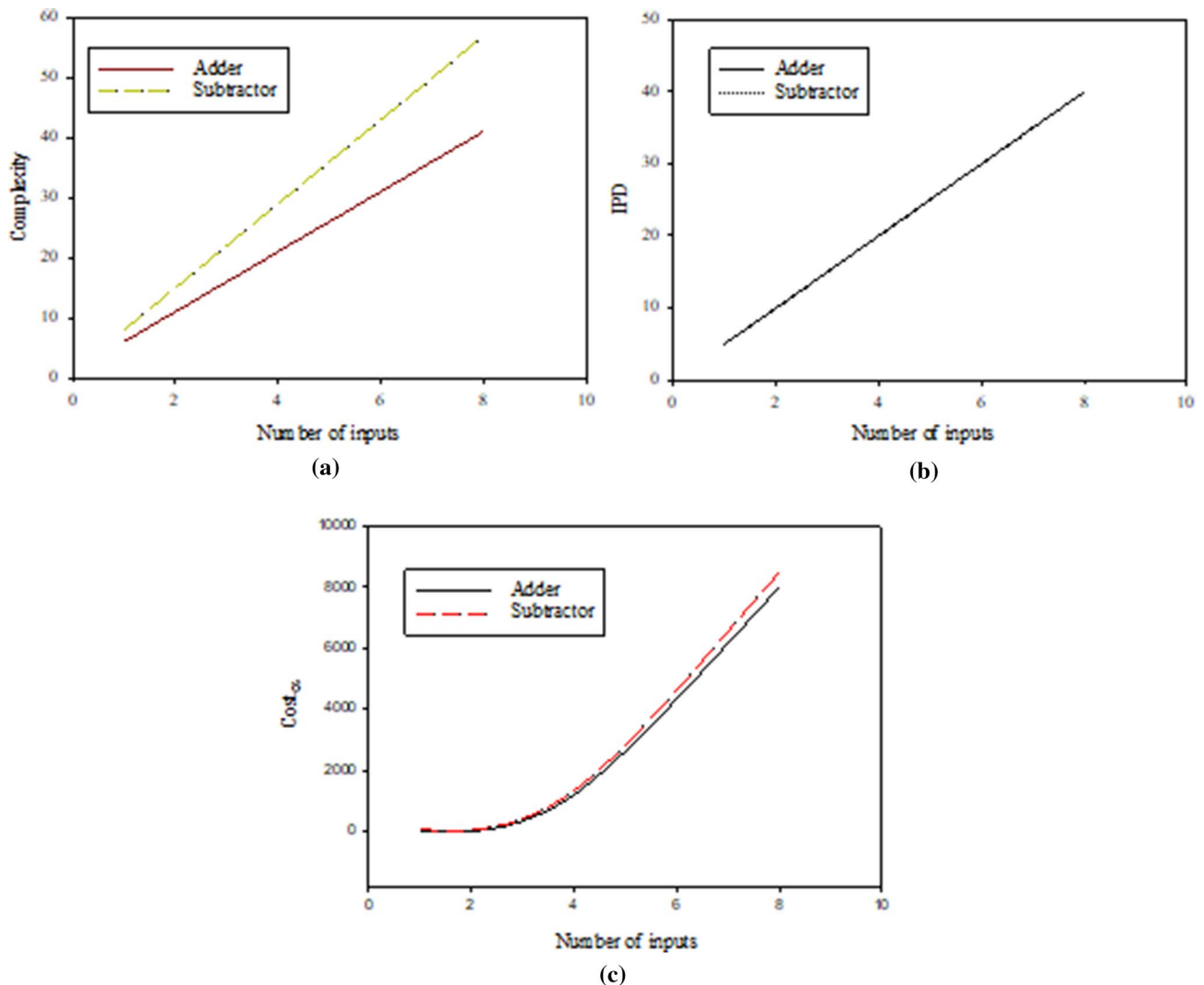
| No | Combinational module | Number of LT gates                                           | Complexity | IPD  | Cost <sub><math>\alpha</math></sub> |
|----|----------------------|--------------------------------------------------------------|------------|------|-------------------------------------|
| 1  | Adder                | LTEx = $2n$<br>LT AND = $2n$<br>LT OR = $n$                  | $5n + 1$   | $5n$ | $(25n^2 + 1) \times T$              |
| 2  | Subtractor           | LTEx = $2n$<br>LT AND = $2n$<br>LT OR = $n$<br>LT NOT = $2n$ | $7n + 1$   | $5n$ | $(25n^2 + 2n + 1) \times T$         |

**Table 8** The calculated value of the LT gates (count), complexity, IPD, and  $Cost_{\alpha}$  for the one-, four-, and eight-bit LT RCA and FS

| No | Combi-national module | Number of LT gates | Complexity | IPD | $Cost_{\alpha}$ |
|----|-----------------------|--------------------|------------|-----|-----------------|
| 1  | Adder                 | One-bit: 5         | 6          | 5   | 39              |
|    |                       | Four-bit: 20       | 21         | 20  | 1203            |
|    |                       | Eight-bit: 40      | 41         | 40  | 8005            |
| 2  | Subtractor            | One-bit: 7         | 8          | 5   | 49              |
|    |                       | Four-bit: 28       | 29         | 20  | 1329.2          |
|    |                       | Eight-bit: 56      | 57         | 40  | 8489.2          |

of a one-bit LT FS needs two LTeX gates, two LT NOT gates, two LT AND gates, and one LT OR gate. The block diagram, LT schematic, and QCA layout of the one-bit LT FS are shown respectively in Fig. 12a, b, and c.

$$\begin{cases} D[0] = LTeX(LTeX(A[0], B[0]), Bin) \\ BO[0] = L_{TO}^-(L_{TA}^+(L_T^-(A[0]), B[0]), L_{TA}^+(Bin, L_T^-(LTeX(A[0], B[0])))) \end{cases} \quad (18)$$

**Fig. 14** A comparative analysis of the proposed adder and subtractor layouts in terms of the **a** complexity, **b** IPD, and **c**  $Cost_{\alpha}$

**Algorithm 4.2: The  $n$ -bit LT FS generation (where  $n=2^m$ ,  $m=0, 1, 2, \dots$ )**

```

INPUT:  $A=(A[0]A[1]\dots A[n-1])$  and  $B=(B[0]B[1]\dots B[n-1])$  in binary, and OUTPUT:  $D=(D[0]D[2]\dots D[n-1])$  and  $BO=(BO[0]BO[2]\dots BO[n-1])$  in binary with initial one-bit borrow input Bin.
1.  $i=0$  //Initialize loop index
2.  $D[0]=\text{LTeX}(\text{LTeX}(A[0],B[0]),\text{Bin});$  //Computation of initial values
    $BO[0]=L_{TO}^-(L_{TA}^+(L_T'(A[0]),B[0]),L_{TA}^+(\text{Bin},L_T'(\text{LTeX}(A[0],B[0]))));$ 
3. for  $i=1$  to  $n-1$  //loop
   do
      $D[i]=\text{LTeX}(\text{LTeX}(A[i],B[i]),BO[i-1]);$ 
      $BO[i]=L_{TO}^-(L_{TA}^+(L_T'(A[i]),B[i]),L_{TA}^+(BO[i-1],L_T'(\text{LTeX}(A[i],B[i]))));$ 
4. end //end of the loop

```

The LT schematic of the one-bit LT FS needs two LTeX gates, two LT NOT gates, two LT AND gates, and one LT OR gate. The block diagram, LT schematic, and QCA layout of the one-bit LT FS are shown respectively in Fig. 12a, b, and c.

The QCA layout of the one-bit LT FS consumes  $0.2 \mu\text{m}^2$  at an O-Cost of 147 and generates the outputs after 1.75 clock cycles. The higher-order iterations of algorithm 2 with  $n=4$  and 8 result in the four- and eight-bit LT FS schematic, respectively. The QCA layouts of the four- and eight-bit LT FS are shown in Fig. 13a and b, respectively.

The one-, four-, and eight-bit LT FS counterparts consume 0.20364, 1.18, and  $3.45 \mu\text{m}^2$  at an O-Cost of 147, 697, and 1661 and generate the outputs after 1.75, 3.25, and 5.25 clock cycles, respectively. As designs are not available in other works for the one- or four-bit FS, we thus compare the eight-bit LT RCA with the eight-bit counterpart designs in Refs. [37, 52] in terms of the effective area in  $\mu\text{m}^2$ , O-Cost, and delay in Table 6.

### 4.3 Cost estimation for the proposed designs

The presence of crossovers as discussed above greatly impacts the QCA design parameters [13, 23, 25]. Existing research shows that logical and multilayer crossovers outperform coplanar counterparts in terms of the layout area, the delay, and the number of cells required. However, multilayer QCA layouts are complex and costly to fabricate. We describe specific QCA design parameters that must be considered to guide the performance of the proposed adder and subtractor layouts and thus understand the foundations of QCA technology better.

The explanation of logic circuits based on the QCA design parameters involves abbreviations of the terms, such as majority gate as M, inverter as I, crossover as C, and delay as T. The individual parameters are described as follows:

- (1) Complexity is the total count of logic gates, inverters (referring to the conventional seven-cell structure shown in Fig. 2e), and crossovers, thus indicating the complexity of the QCA layout, i.e.,  $M+I+C$ .

- (2) The irreversible power dissipation (IPD) is the count of the logic gates directly involved, i.e., M.

- (3)  $\text{Cost}_\alpha$ , i.e., the QCA cost function, is expressed by the mathematical expression in Eq. (19) [13].

$$\text{Cost}_\alpha = (M^2 + I + C^2) \times T \quad (19)$$

The number of LT gates (count), the complexity, the IPD, and the  $\text{Cost}_\alpha$  of the generic adder and subtractor layouts are precisely modeled in Table 7.

The number of LT gates (count), complexity, IPD, and  $\text{Cost}_\alpha$  of the one-, four-, and eight-bit LT RCA and FS are calculated and presented in Table 8.

Figure 14 illustrates the complexity, IPD, and  $\text{Cost}_\alpha$  for the proposed adder and subtractor layouts with increasing number of input bits. The variation of the complexity (Fig. 14a) shows higher values for the subtractors than adders. The IPD of the adders and subtractors remains the same, as demonstrated in Fig. 14b. The adder's  $\text{Cost}_\alpha$  curve shows a lower variance than that of the subtractor, which is due to the additional inverter in the LT subtractor layouts. The variation of  $\text{Cost}_\alpha$  is shown in Fig. 14c.

### 4.4 Result analysis

The LT cell library facilitates the generation of QCA layouts for a generic adder and subtractor. Algorithm 1 is proposed to instantiate an  $n$ -bit LT RCA. The one-, four-, and eight- LT RCAs are generated as instances of the LT RCA counterparts. The results presented in Table 8 show that the proposed one-bit LT RCA consumes 25.6% less effective area than the existing counterpart in Ref. [30] with 31.4% lower O-Cost than the RCA counterpart in Ref. [31]. The four-bit LT RCA consumes 4.33% less effective area and has 33.33% lower O-Cost than the existing counterpart [36]. Although the eight-bit LT RCA design consumes more effective area than the RCA in Ref. [36], the O-Cost requirement shows a 29.5% improvement over the eight-bit design [36]. Other available systems can implement RCAs with lower effective area and O-Cost, but they do not obey the QCA design rules properly.

Further, algorithm 2 is introduced to generate  $n$ -bit LT FS. The one-, four-, and eight-bit LT FSs are instantiated by utilizing the proposed algorithm. The eight-bit LT FS is compared with the existing counterpart in Ref. [52]. The eight-bit LT FS design requires a 11.84% lower O-Cost to generate the output 81.25% faster in comparison with the eight-bit FS presented in Ref. [52]. Tables 10, 11, and Fig. 14 illustrate the analytical behavior of the  $n$ -bit LT RCA and FS in terms of QCA design parameters such as the complexity, the IPD, and the Cost<sub>α</sub>.

## 5 Conclusions

An entire LT cell library is created and further utilized to construct lower- and higher-order adder and subtractor modules. As the average area requirement of LT cells is lower than that of their MV counterparts, the LT logic reduction methodology emerges as a suitable alternative to the conventional MV logic reduction technique. A comparative analysis with existing adder and subtractor designs aids in the identification of the drawbacks and challenges of prevailing works. Two algorithms are proposed to create  $n$ -bit LT RCA and LT FS QCA layouts. Structures that overcome the disadvantages of existing designs are generated. The cost is estimated for the  $n$ -bit LT RCA and FS designs. The comparative analysis presented herein contributes significantly to the advancement of QCA-based nanotechnology research, specifically nano communication network and error control nano-encoder designs.

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